Final Exam

Graded

Student

Leyan Wang

Total Points

42.5 / 45 pts

Question 1

Overall Grades **42.5** / 45 pts

✔ **– 0 pts** See the details on each problem.

💬 **– 2.5 pts** Point adjustment

MATH 202–001 FALL 2026 SESSION 1 FINAL EXAM
OCTOBER 19TH, 15:30–18:30

Name (Please Print)	NetID	Signature
Leyan Wang.	lw455.	王乐彦.

INSTRUCTION

- This is a 180-minute test.
- The total number of points is 45.
- No Cellphones.
- No Calculators.
- Do NOT write on the back of the test; if you do so, your answer(s) will NOT be graded.
- There are 9 problems.
- Show your work. To receive full credit, your answers must be neatly written, logically organized, and clearly labeled. Writing just the answer will not give you credit for that.
- The backs and extra pages are offered as scratch paper.
- GOOD LUCK.

Duke Community Standard

Duke Kunshan University (DKU) is a community comprised of individuals from diverse cultures and backgrounds. We are dedicated to scholarship, leadership, and service and to the principles of honesty, fairness, respect, and accountability. Members of this community commit to reflecting upon and upholding these principles in all academic and non-academic endeavors and to protecting and promoting a culture of integrity and trust.

To uphold the DKU Community Standard

- I will hold myself to the highest standards for honesty, integrity, fairness, and responsibility in my academic and non-academic endeavors
- I will respect other cultures and embrace all forms of diversity
- I will uphold the standards if they are compromised.

Honor Pledge: I will neither give nor receive aid on this test. SIGNATURE: 王乐彦.

IF YOU DO NOT SIGN THE HONOR PLEDGE, YOUR FINAL EXAM WILL BE RECORDED AS 0 POINTS.

Problems	1	2	3	4	5	6	7	8	9	Total
Scores	5	4.5	5	5	3	5	5	5	5	42.5

Problem 1. [5] Find the general solution of the linear system:

$$2x - 3y = 1$$
$$-2y - 3z = 1$$
$$x + 2z = 1$$

First write into a matrix :

$$\begin{bmatrix} 2 & -3 & 0 \\ 0 & -2 & -3 \\ 1 & 0 & 2 \end{bmatrix}$$

$R_3 \to 2R_3 - R_1 = \begin{bmatrix} 2 & -3 & 0 \\ 0 & -2 & -3 \\ 0 & 3 & 4 \end{bmatrix}$; $R_3 \to 2R_3 + 3R_2$

Write B : Homogeneous Matrix

$$= \begin{bmatrix} 2 & -3 & 0 \\ 0 & -2 & -3 \\ 0 & 0 & -1 \end{bmatrix}$$

Or just solve it :

$$\begin{cases} 2x - 3y = 1 & (1) \\ -2y - 3z = 1 & (2) \\ x + 2z = 1 & (3) \end{cases}$$

(4)

(3) → (1) − 2*(3) = $-3y - 4z = -1$

→ (2) − (4) : $y + z = 2$

(3) + 2*(5) : $z = -5$ $y = 7$

$x = 11$

$-z = 1$. $z = -1$.

$-2y - 3z = 1$. $-2y = -2$

$y = 1$

$2x - 3y = 1$

$x = 2$.

Because :

$n = r$ So there is consistent ~~inconsistent~~ limited solution which is :

Ans :

$$\begin{cases} x = 11 \\ y = 7 \\ z = -5 \end{cases}$$

4.5/5

Problem 2. [5'] Let $A = \begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$. Find

(1) a basis of the column space of A,
(2) a basis of the row space of A,
(3) and a basis of the null space of A.

(1). $\begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$ $R_2 \to R_2 + 2R_1 + R_4$, $R_4 \to R_4 + R_2 + 2R_1$ $=$

$\begin{bmatrix} 2 & 4 & 2 \\ 0 & 11 & 11 \\ 0 & 1 & 1 \\ 0 & 11 & 11 \end{bmatrix} = \begin{bmatrix} 2 & 4 & 2 \\ 0 & 11 & 11 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$.

Basis for (1) $= Col(A) = \left\{ \begin{bmatrix} 2 \\ -5 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ 0 \\ 1 \\ 3 \end{bmatrix} \right\}$

Basis for (2) $= Row(A) = \left\{ \begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$ why.

-0.5

(3): take $z = 5$, $y + 5 = 0$ $y = -5$; $2x - 45 + 25 = 0$

$x = 5$.

$A = 5 \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$ So $nullity(A) = n - r = 1$.

Basis for it $= \left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}$

5/5

Problem 3. [5'] Let $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^2$ be a linear transformation such that

$$T\left(\begin{bmatrix}0\\1\\-1\end{bmatrix}\right) = \begin{bmatrix}1\\1\end{bmatrix}, \quad T\left(\begin{bmatrix}1\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}0\\2\end{bmatrix}, \quad T\left(\begin{bmatrix}0\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}2\\2\end{bmatrix}.$$

Find the matrix transformation of T, i.e., find a matrix A such that

$$T\left(\begin{bmatrix}x\\y\\z\end{bmatrix}\right) = A\begin{bmatrix}x\\y\\z\end{bmatrix}.$$

We would use $T(v) = Av$,

So $\begin{cases} A\begin{bmatrix}0\\1\\-1\end{bmatrix} = \begin{bmatrix}1\\1\end{bmatrix} \\ A\begin{bmatrix}1\\0\\-1\end{bmatrix} = \begin{bmatrix}0\\2\end{bmatrix} \\ A\begin{bmatrix}0\\0\\-1\end{bmatrix} = \begin{bmatrix}2\\2\end{bmatrix} \end{cases}$

A should be: 2×3.

We also know for $u = av_1 + bv_2$

$$T(u) = aT(v_1) + bT(v_2)$$

We create $u_1 = \begin{bmatrix}0\\1\\0\end{bmatrix} = v_1 - v_3$ ✓

$u_2 = \begin{bmatrix}1\\0\\0\end{bmatrix} = v_2 - v_3$

and $u_3 = \begin{bmatrix}0\\0\\1\end{bmatrix} = -v_3$.

So $\begin{cases} A\cdot\begin{bmatrix}0\\1\\0\end{bmatrix} = \begin{bmatrix}-1\\-1\end{bmatrix} \\ A\begin{bmatrix}1\\0\\0\end{bmatrix} = \begin{bmatrix}-2\\0\end{bmatrix} \\ A\begin{bmatrix}0\\0\\1\end{bmatrix} = \begin{bmatrix}-2\\-2\end{bmatrix} \end{cases}$

So A would be $\begin{bmatrix}-2 & -1 & -2\\ 0 & -1 & -2\end{bmatrix}$

Examine: Av_1, Av_2, Av_3 all correct!

Ans: $A = \begin{bmatrix}-2 & -1 & -2\\ 0 & -1 & -2\end{bmatrix}$

Problem 4. [5] Find the Jordan canonical form of

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ -2 & -2 & 1 \end{bmatrix}$$

First we need engine values λ

Solve: let $p(\lambda) = |A - I\lambda| = 0$ $\quad A - I\lambda = \begin{bmatrix} 1-\lambda & 2 & 0 \\ 0 & 2-\lambda & 0 \\ -2 & -2 & 1-\lambda \end{bmatrix}$

$\det(A - I\lambda) = 0 \cdot C_{21} + (2-\lambda) C_{22} + 0 \cdot C_{23}$

$= (2-\lambda) \cdot M_{22} = (2-\lambda)[(1-\lambda)^2 - 0]$

$= (2-\lambda)(2-\lambda)(1-\lambda)$

$\lambda_1 = 2 \quad \lambda_2 / \lambda_3 = 1.$

① For $\lambda_1 = 2$. $\quad A - I\lambda = \begin{bmatrix} -1 & 2 & 0 \\ 0 & 0 & 0 \\ -2 & -2 & -1 \end{bmatrix}$ $\quad v_3 \begin{bmatrix} 1 \\ \frac{1}{2} \\ -3 \end{bmatrix}$.

② For $\lambda_2 / \lambda_3 = 1$. $\quad A - I\lambda = \begin{bmatrix} 0 & 2 & 0 \\ 0 & 1 & 0 \\ -2 & -2 & 0 \end{bmatrix}$.

~~Algebra~~ multiplicity { Algibrac = 1. Geometric = 1.

Using $[P|I] = [I|P^{-1}]$ $= \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

We get $P^{-1} = \begin{bmatrix} -7 & -8 & -3 \\ 2 & -2 & 0 \\ 6 & -6 & 3 \end{bmatrix}$ multiply

Algibrac = 2. $\quad x = -y$.

Geometric = 1. $\quad v_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$ — OF

This is wrong but it doesn't matter

We need to find the generalized engine vector

$(A - I\lambda) v_2 = v_1 \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \cdot v_2 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$

Ans: J would be: $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ with $\begin{bmatrix} J_2 & 0 \\ 0 & J_1 \end{bmatrix}$ v_2 can be: $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

Corresponding P is $\begin{bmatrix} 1 & 1 & 1 \\ -1 & 1 & \frac{1}{2} \\ 0 & 1 & -3 \end{bmatrix}$ $\quad A = PJP^{-1}$

3/5

Problem 5. [5'] Diagonalize the quadratic form

$$f(x, y, z) = 3x^2 + 3y^2 + 4z^2 + 2xz - 2yz.$$

First write this quadratic form into matrix

Which is: $\begin{bmatrix} 3 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix}$ then find λ for it.

$p(\lambda) = 0$ $\det\left(\begin{bmatrix} 3-\lambda & 0 & 1 \\ 0 & 3-\lambda & -1 \\ 1 & -1 & 4-\lambda \end{bmatrix}\right)$ $(3-\lambda)$

$= (3-\lambda)C_{11} + 0 + 1 \cdot C_{13}$

$= (3-\lambda)\left[(3-\lambda)(4-\lambda) - 1\right] + 1 \cdot (\lambda - 3)$

$= (3-\lambda)\left[\lambda^2 - 7\lambda + 10\right]$

$(3-\lambda)(\lambda-2)(\lambda-5)$

There're 3 λ so their v should be linear independent

$\lambda = 3$

$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & -1 \\ 1 & -1 & 1 \end{bmatrix}$ $v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ -0.5

$\lambda = 2$

$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & -1 & 2 \end{bmatrix}$ $v_2 = \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}$

$\lambda = 5$

$\begin{bmatrix} -2 & 0 & 1 \\ 0 & -2 & -1 \\ 1 & -1 & -1 \end{bmatrix}$ $v_3 = \begin{bmatrix} 1 \\ 2 \\ -4 \end{bmatrix}$ -0.5

D can be $\begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{bmatrix}$.

Ans: $f(x, y, z) = f(u, v, w)$

$= 3u^2 + 2v^2 + 5w^2$

$P = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 1 & 2 \\ 0 & -1 & -4 \end{bmatrix}$.

$A = PDP^{-1}$.

Problem 6. [5'] Find a, b, and c such that $\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$, where

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \quad \text{and} \quad \mathbf{w} = \begin{bmatrix} -3 \\ 8 \\ -8 \end{bmatrix}.$$

We can use the ~~Aug~~ Augmented Matrix to Solve.

$$\left[\begin{array}{ccc|c} 0 & 2 & -1 & -3 \\ 2 & -1 & 1 & 8 \\ -2 & 2 & 0 & -8 \end{array}\right] \text{ We write } [v_1\ v_2\ v_3] \text{ As } A.$$

$Ax = b$ So $x = A^{-1}b$ *

$[A|I] = [I|A^{-1}]$

$$[A|I] = \left[\begin{array}{ccc|ccc} 0 & 2 & -1 & 1 & 0 & 0 \\ 2 & -1 & 1 & 0 & 1 & 0 \\ -2 & 2 & 0 & 0 & 0 & 1 \end{array}\right]$$

$R_2 \to R_2 + R_3 =$

$$\left[\begin{array}{ccc|ccc} 0 & 2 & -1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 \\ -2 & 2 & 0 & 0 & 0 & 1 \end{array}\right]$$

$R_1 \to \frac{R_1+R_2}{2}$:

$$\left[\begin{array}{ccc|ccc} 0 & 1 & 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ 0 & 1 & 1 & 0 & 1 & 1 \\ -2 & 2 & 0 & 0 & 0 & 1 \end{array}\right]$$

$\begin{cases} R_2 \to R_2 - R_1 \\ R_3 \to -\frac{R_2}{2} + R_1 \end{cases}$:

$$\left[\begin{array}{ccc|ccc} 0 & 1 & 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ 0 & 0 & 1 & -\frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ 1 & 0 & 0 & \frac{1}{3} & \frac{1}{3} & -\frac{1}{6} \end{array}\right]$$

$$= \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{3} & \frac{1}{3} & -\frac{1}{6} \\ 0 & 1 & 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ 0 & 0 & 1 & -\frac{1}{3} & \frac{2}{3} & \frac{2}{3} \end{array}\right] \quad A^{-1}$$

$x = A^{-1}b = \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix}$

Ans: $w = 3v_1 - v_2 + v_3$

Problem 7. [5] Let V be a subspace of $\mathbb{R}^4$, defined by

$$V = span\left\{ \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ 2 \\ -1 \end{bmatrix} \right\}.$$

Find an orthonormal basis for V.

v_1 v_2

First of All, we can see obviously v_1 & v_2 are linear independent which means, it's a plane in R^4.

So (Step 1) is to make them orthogonal is to use "Gram - Schmidt Orthogonalization"

Formula: $v_1 - \frac{v_1 \cdot v_2}{\|v_2\|^2} \cdot v_2$ is $\begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} - \frac{2}{14} \cdot \begin{bmatrix} 3 \\ 0 \\ 2 \\ -1 \end{bmatrix}$

$\|v_2\| = \sqrt{9+4+1} = \sqrt{14}$ $= \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} \frac{3}{7} \\ 0 \\ \frac{2}{7} \\ -\frac{1}{7} \end{bmatrix}$

* Examine $v_2 \cdot v_3 = -\frac{9}{7} + \frac{10}{7} - \frac{1}{7} = 0$ (✓) $= \begin{bmatrix} -\frac{3}{7} \\ 1 \\ \frac{5}{7} \\ \frac{1}{7} \end{bmatrix} = v_3$.

Now we need to normalize them

Change v_3 to $\begin{bmatrix} -3 \\ 7 \\ 5 \\ 1 \end{bmatrix}$ (doesn't change direction but norm)

$\|v_3\| = \sqrt{9+49+25+1} = \sqrt{84} = 2\sqrt{21}$

$\frac{v_2}{\|v_2\|} = \begin{bmatrix} \frac{3}{\sqrt{14}} \\ 0 \\ \frac{2}{\sqrt{14}} \\ \frac{-1}{\sqrt{14}} \end{bmatrix}$ $\frac{v_3}{\|v_3\|} = \begin{bmatrix} \frac{-3}{2\sqrt{21}} \\ \frac{7}{2\sqrt{21}} \\ \frac{5}{2\sqrt{21}} \\ \frac{1}{2\sqrt{21}} \end{bmatrix}$

$\hookrightarrow v_2'$ $\hookrightarrow v_3'$

Ans:

Orthonormal Basis for V is $\{v_2', v_3'\}$

5/5

Problem 8. [5] Let A be an $n \times n$ square matrix satisfying $A^T = -A$. Show that if n is odd, then $\det A = 0$.

A is skew-symmetric,

We know that $\det(A^T) = \det(A)$

&

$\det(kA) = k^n \det(A)$ take $k = -1$.

$\det(-A) = (-1)^n \det(A)$

$(-1)^n \det(A) = \det(A)$ when $n = 2k+1$ $(k \in \mathbb{Z})$

Only way to make $\det(A) = -\det(A)$ is $\det(A) = 0$.

5/5

Problem 9. [5'] For two vectors $\vec{u}$ and $\vec{v}$, suppose they satisfy that

$$\|\vec{u}+\vec{v}\| = \|\vec{u}-\vec{v}\|.$$

Show that $\vec{u}$ and $\vec{v}$ are perpendicular to each other.

Simply do ~~$(\vec{u}+\vec{v})^2$~~ #

$(\|\vec{u}+\vec{v}\|)^2$ ⊘ $(\|\vec{u}-\vec{v}\|)^2$

as we know $\|\vec{a}\|$ is always ≥ 0

~~So it can't be $\|\vec{u}+\vec{v}\| = -\|\vec{u}-\vec{v}\|$~~

*. the only possibility for $\|\vec{u}+\vec{v}\| = -\|\vec{u}-\vec{v}\|$ is $\vec{u} = \vec{v} = 0$, in this case, they're ~~pep~~ perpendicular!

Otherwise:

$(\vec{u}+\vec{v})^2 = (\vec{u}-\vec{v})^2 \Leftrightarrow \|\vec{u}+\vec{v}\| = \|\vec{u}-\vec{v}\|$

In this case

$\vec{u}^2 + \vec{v}^2 + 2\vec{u}\cdot\vec{v} = \vec{u}^2 + \vec{v}^2 - 2\vec{u}\cdot\vec{v}$

$4\vec{u}\cdot\vec{v} = 0$

$\vec{u}\cdot\vec{v} = 0$.

& here $\|\vec{u}\| \neq 0, \|\vec{v}\| \neq 0$.

$\cos\theta = 0$, $\theta = 90^\circ$!

~~A~~